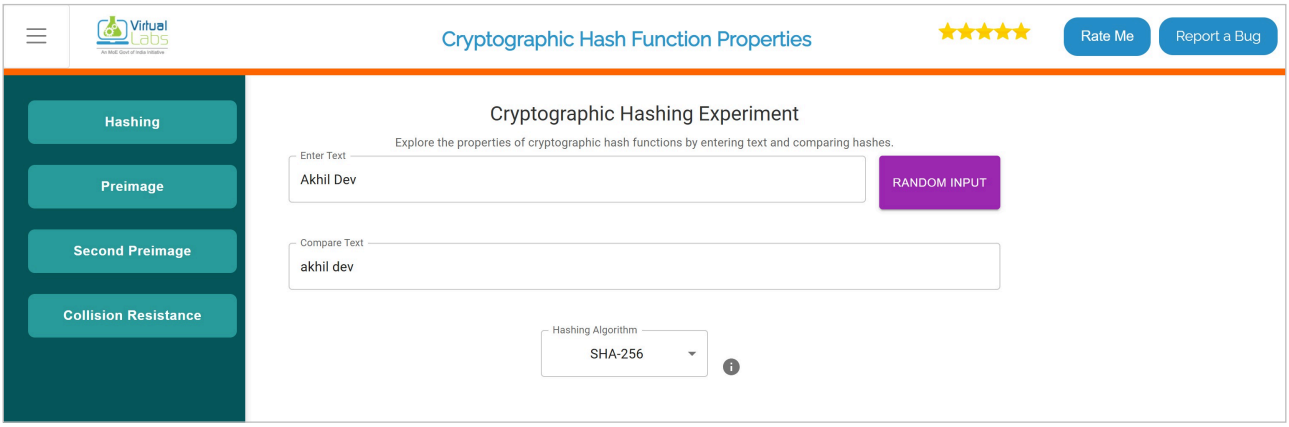
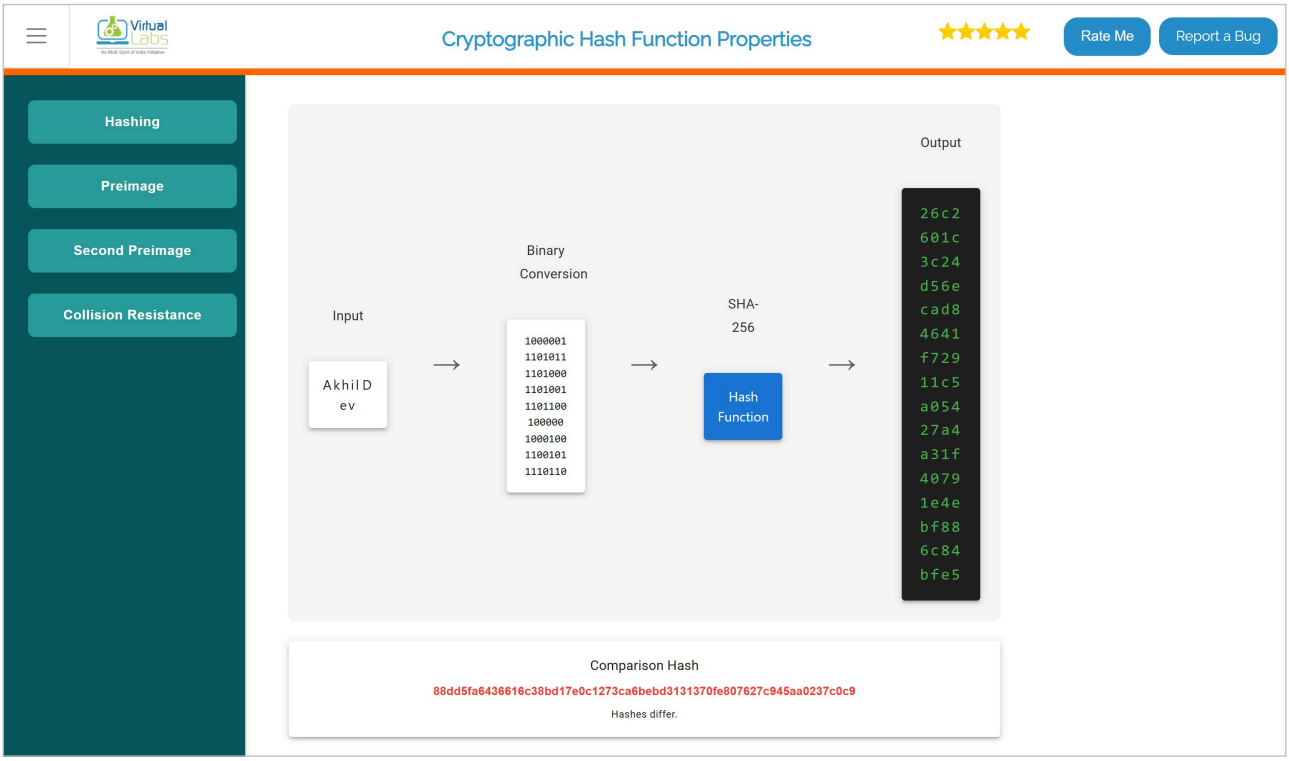
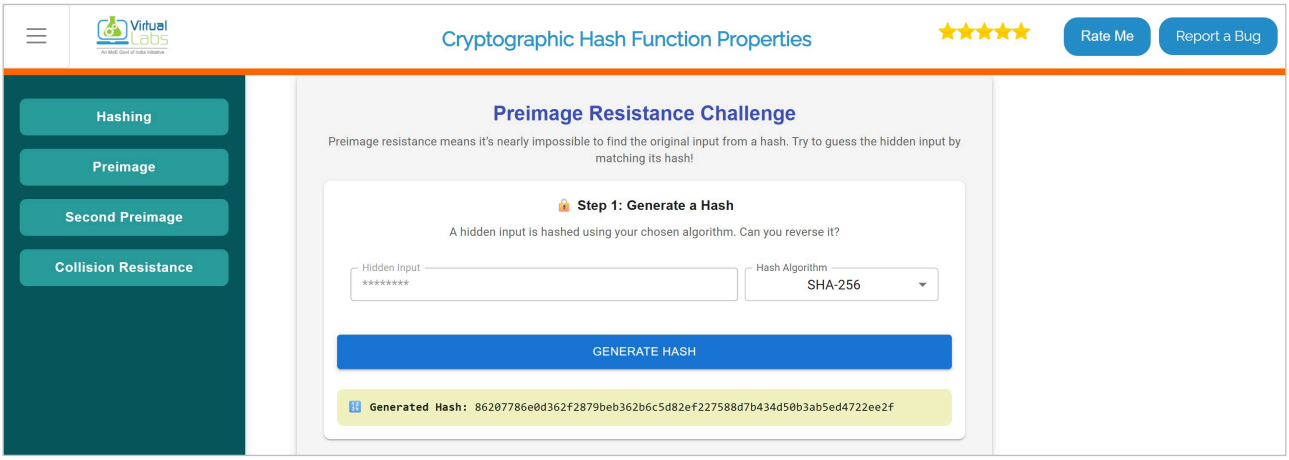
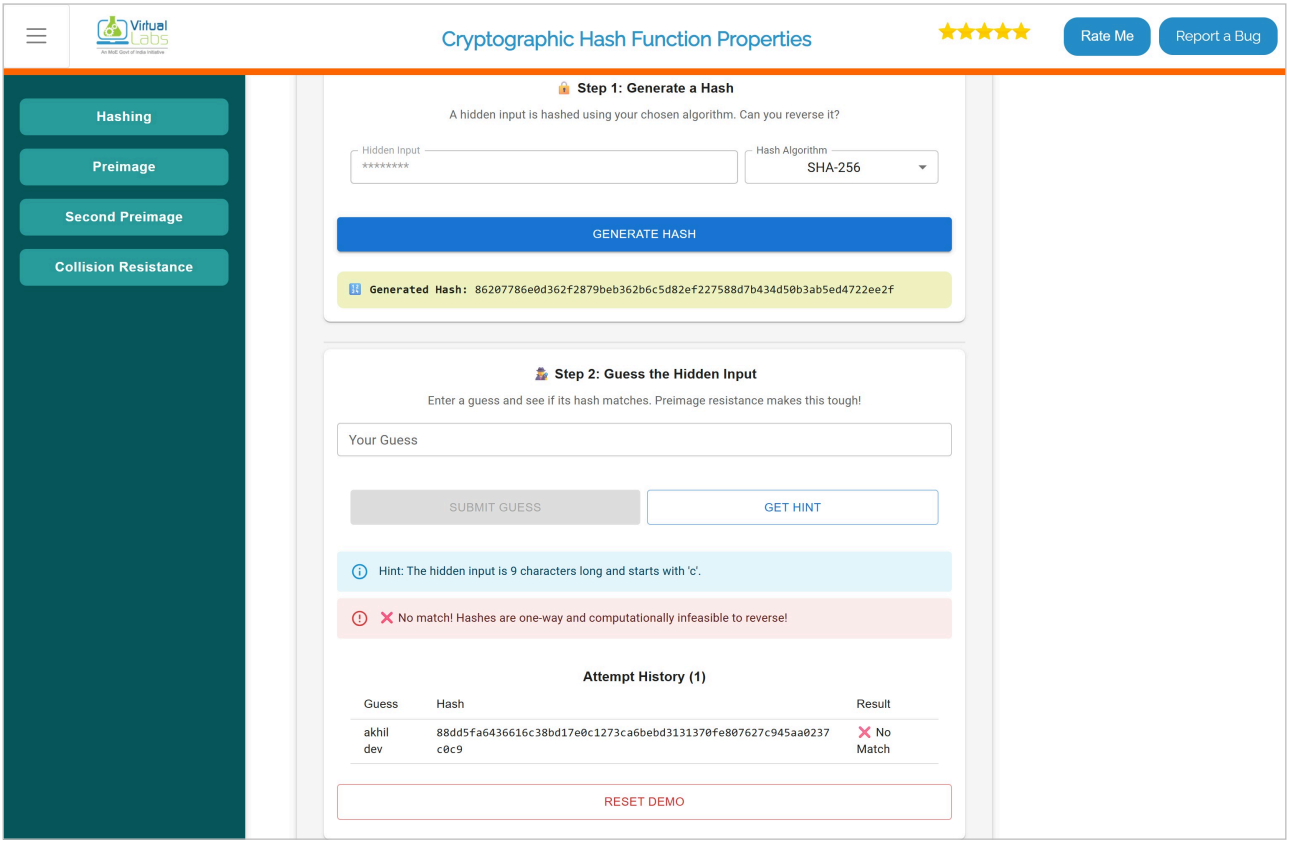
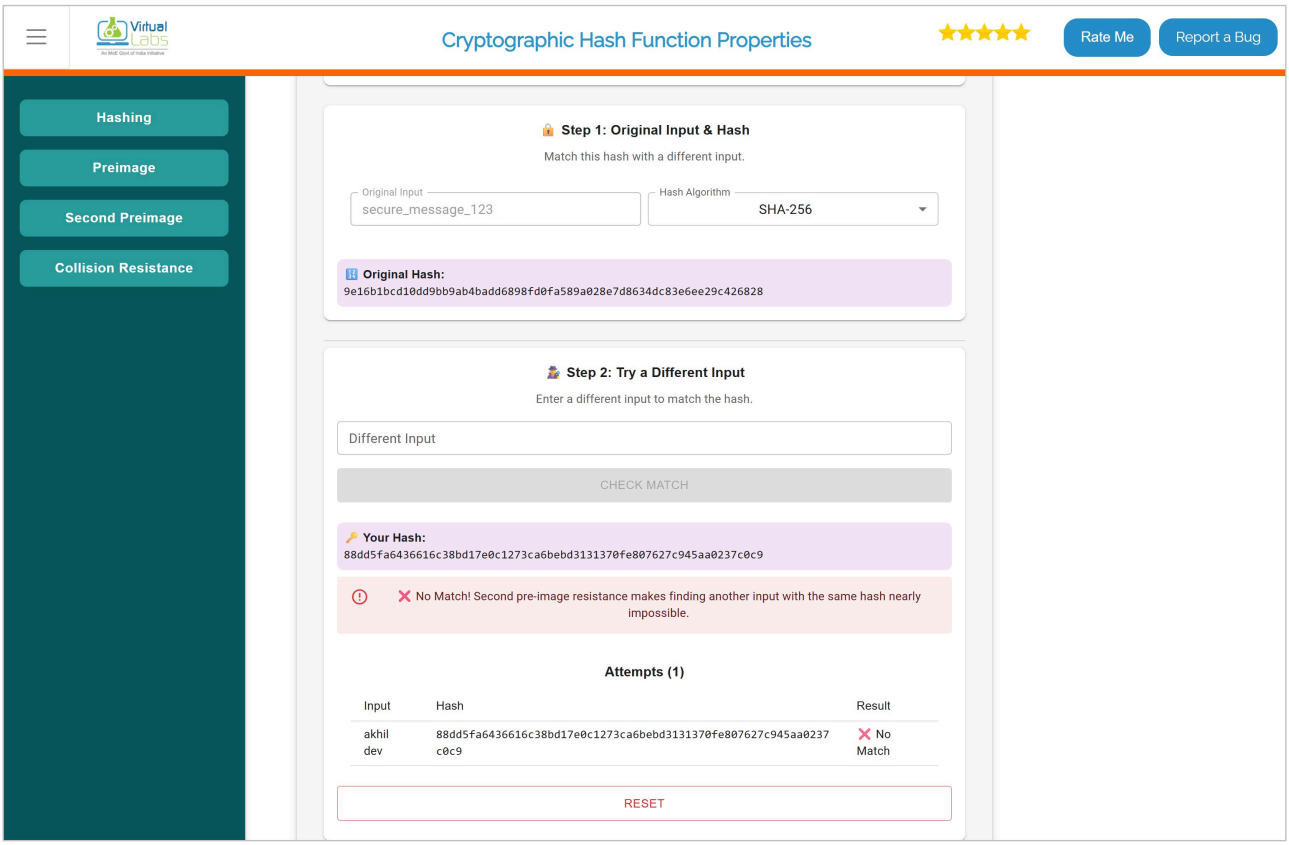
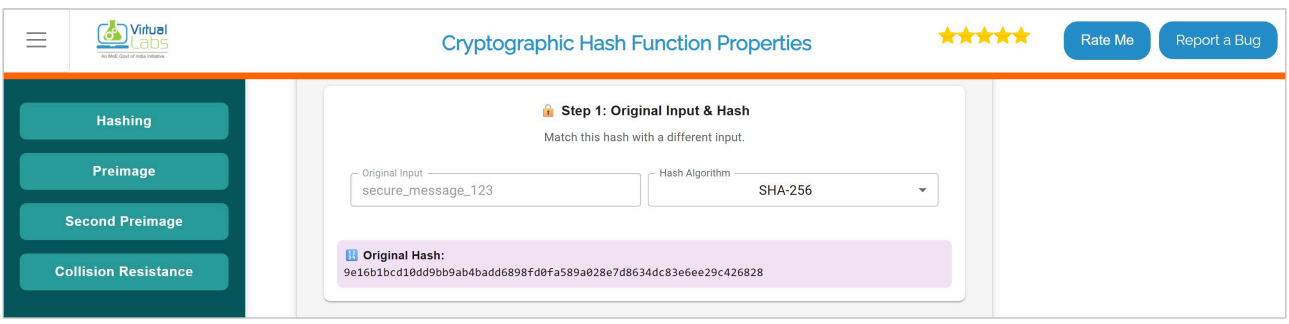

Virtual Labs
Cryptographic Hash Function Properties
Rate Me
Report a Bug
Hashing
Preimage
Second Preimage
Collision Resistance
Cryptographic Hashing Experiment
Explore the properties of cryptographic hash functions by entering text and comparing hashes.
Enter Text
Akhil Dev
RANDOM INPUT
Compare Text
akhil dev
Hashing Algorithm
SHA-256


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Collision Resistance
Input
Akhil Dev
Binary Conversion
1000001
1101011
1101000
1101001
1101100
100000
1000100
1100101
1110110
SHA-256
Hash Function
Output
26c2
601c
3c24
d56e
cad8
4641
f729
11c5
a054
27a4
a31f
4079
1e4e
bf88
6c84
bfe5
Comparison Hash
88dd5fa6436616c38bd17e0c1273ca6bebd3131370fe807627c945aa0237c0c9
Hashes differ.


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Preimage Resistance Challenge
Preimage resistance means it's nearly impossible to find the original input from a hash. Try to guess the hidden input by matching its hash!
Step 1: Generate a Hash
A hidden input is hashed using your chosen algorithm. Can you reverse it?
Hidden Input
****
Hash Algorithm
SHA-256
GENERATE HASH
Generated Hash: 86207786e0d362f2879beb362b6c5d82ef227588d7b434d50b3ab5ed4722ee2f


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Step 1: Generate a Hash
A hidden input is hashed using your chosen algorithm. Can you reverse it?
Hidden Input
****
Hash Algorithm
SHA-256
GENERATE HASH
Generated Hash: 86207786e0d362f2879beb362b6c5d82ef227588d7b434d50b3ab5ed4722ee2f
Step 2: Guess the Hidden Input
Enter a guess and see if its hash matches. Preimage resistance makes this tough!
Your Guess
SUBMIT GUESS
GET HINT
Hint: The hidden input is 9 characters long and starts with 'c'.
No match! Hashes are one-way and computationally infeasible to reverse!
Attempt History (1)
Guess
Hash
Result
akhil dev
88dd5fa6436616c38bd17e0c1273ca6bebd3131370fe807627c945aa0237c0c9
No Match
RESET DEMO


Virtual Labs
Cryptographic Hash Function Properties
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Second Preimage
Collision Resistance
Step 1: Original Input & Hash
Match this hash with a different input.
Original Input
secure_message_123
Hash Algorithm
SHA-256
Original Hash:
9e16b1bcd10dd9bb9ab4badd6898fd0fa589a028e7d8634dc83e6ee29c426828
Step 2: Try a Different Input
Enter a different input to match the hash.
Different Input
CHECK MATCH
Your Hash:
88dd5fa6436616c38bd17e0c1273ca6bebd3131370fe807627c945aa0237c0c9
No Match! Second pre-image resistance makes finding another input with the same hash nearly impossible.
Attempts (1)
Input
Hash
Result
akhil dev
88dd5fa6436616c38bd17e0c1273ca6bebd3131370fe807627c945aa0237c0c9
No Match
RESET


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Cryptographic Hash Function Properties
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Step 1: Original Input & Hash
Match this hash with a different input.
Original Input
secure_message_123
Hash Algorithm
SHA-256
Original Hash:
9e16b1bcd10dd9bb9ab4badd6898fd0fa589a028e7d8634dc83e6ee29c426828